

Learned Spacecraft Pose Estimation: Failure Detection and Uncertainty

Experiments on monocular relative pose estimation for rendezvous and proximity operations

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1. Problem statement

Autonomous rendezvous and docking relies on active sensing and cooperative targets. Learned monocular pose estimation is proposed in multiple research papers as a way to extend that capability to unprepared targets and degraded illumination, but it is not yet flown. Three barriers are identified consistently in the literature review of multiple papers: the synthetic-to-real domain gap, deployment inside a flight processor and its budget, and predictable behaviour when an estimate is unreliable.

The following experiments explore and approach the 3 problems mentioned above. The question throughout is, not which model is most accurate, but **whether the perception stage can determine that it is wrong, at runtime and without ground truth**. A pose returned with high confidence that is in fact wrong, propagates unchecked into the navigation filter. A pose flagged as uncertain can be rejected, retried or cross-checked against another sensor. Accuracy and self-assessment are therefore separate requirements, and this assessment is the one that decides whether a learned component can sit inside an engineering navigation architecture.

2. Experiments performed

S/N	Title	Data	Principal source
1	Keypoint detector benchmark on clean imagery (ORB, SIFT, DISK)	Synthetic wireframe	Kisantal et al. 2020 [3]; Tyszkiewicz et al. 2020 [7]
2	Robustness under controlled degradation	Synthetic wireframe	Pauly et al. 2023 [6]
3	Failure detection and confidence calibration	Synthetic wireframe	Guo et al. 2017 [8]
4	Convolutional and transformer backbones; measurement of the domain gap	SPEED+ v2	Park et al. 2022 [4]; Oquab et al. 2024 [9]
5	Aleatoric and epistemic uncertainty across the domain gap	SPEED+ v2	Kendall & Gal 2017 [10]; Gal & Ghahramani 2016 [11]
6	Confidence-driven measurement covariance in an EKF; Monte Carlo campaign	Simulated HCW trajectory	Martinez et al. 2023 [1]
7	Deployment cost: parameters, operations, latency, quantisation	Both data models	Cosmas & Asami 2020 [12]; Posso et al. 2024 [13]

Experiments 1 to 3 share one notebook and one testbed. Experiments 4 and 5 share the SPEED+ data and the same network. Experiment 6 is the filter study, Experiment 7 the deployment study. All code and executed notebooks are in the accompanying repository.

3. Experiments 1 - 3: detector benchmark, robustness and calibration

Idea: The hybrid architecture used here, is inspired from the solutions developed in first Satellite Pose Estimation Competition 2020 [3]. A network locates known structural keypoints, and a Perspective-n-Point solver with RANSAC recovers the pose geometrically. That separation keeps the geometry analysable. The experiment's focus is to answer, how the choice of keypoint detector propagates through it, and whether the solver's own by-product, the RANSAC inlier ratio, can serve as a runtime confidence signal.

Method. A programmatic spacecraft-like wireframe with 16 keypoints and exact ground truth, rendered at 40 random poses. Three detectors: ORB and SIFT as classical baselines, and DISK [7], a pretrained convolutional detector and descriptor, as the learned component. Four degradations (motion blur, brightness, additive noise, occlusion) at six severity levels, giving 2,880 runs. Failure is defined as translation error above 1 m or rotation above 10°. Calibration is measured with Expected Calibration Error [8].

Detector	Detections	Matched (of 16)	Transl. (m)	Rot. (°)	ECE	ms/frame
ORB	585	16	0.092	0.77	0.091	6.2
SIFT	44	7	0.176	4.57	0.188	73.7
DISK (CNN)	1024	9	0.232	0.50	0.232	72.7

Table 1. accuracy, calibration and timing. Colab T4 gpu is used here; DISK has 1.09 M parameters.

Result. Detection count does not predict usefulness: DISK produced the most detections and the fewest that landed on model keypoints. Accuracy and calibration are separate. DISK produced the best clean rotation accuracy and the worst-calibrated confidence, ECE 0.232 against 0.091 for ORB. When its inlier ratio reported high confidence, the pose was wrong considerably more often than that value implied. The degradations separated the detectors in physically explainable ways, motion blur degraded ORB first, since FAST corner response depends on high-frequency edge energy. But occlusion beyond about one-third of the target flattened(negatively affected) all three detectors together.

Limitation. Association in these experiments used the ground-truth projections of the model keypoints. This is a reasonable model of a cooperative target carrying identified markers, as in the InSPoC-1 architecture [1], but not of an unprepared one, and it makes absolute figures optimistic. The comparison between detectors is unaffected.

4. Experiment 4: backbone comparison and measurement of the domain gap

Idea: Experiments 1 to 3 used only rendered imagery, so they did not measure the distance between rendered and real images. SPEED+ [4] dataset was used for this experiment: 59,960 labelled synthetic images of the Tango spacecraft, plus two hardware-in-the-loop domains photographed in Stanford SLAB's TRON facility; lightbox, simulating Earth albedo, and sunlamp, simulating direct sunlight. The SPEED+ v2 dataset includes labels for both, so the Domain gap can be measured rather than inferred. The second question is whether a transformer backbone helps, since attention is one of the directions under consideration.



Figure 1: the visual difference between the three domains: evenly lit synthetic images, Earth in frame on lightbox, saturation and stray light on sunlamp

Method. Two backbones are tested under an identical head, loss and schedule. A fully fine-tuned ResNet-18 (11.57 M parameters, all trained) and a frozen DINOv2 ViT-S/14 [9] (22.58 M parameters, 0.53 M trained in the head). 8,000 synthetic training images, 12 epochs, evaluated on 1,500 held-out synthetic and 600 images per hardware-in-the-loop domain. No domain adaptation was applied, deliberately, so that the untreated gap is what is observed.

Model	Domain	Transl. (m)	% of range	Rot. (°)	Factor
ResNet-18	synthetic	0.271	5.07	27.5	1.00
ResNet-18	lightbox	3.920	64.19	133.4	14.47
ResNet-18	sunlamp	4.174	66.64	132.8	15.41
DINOv2 ViT-S/14	synthetic	0.366	6.33	52.9	1.00
DINOv2 ViT-S/14	lightbox	2.100	36.26	107.9	5.74
DINOv2 ViT-S/14	sunlamp	1.591	26.46	106.9	4.35

Table 2. Median pose error by domain. 'Factor' is degradation relative to the synthetic domain.

Result. The ranking reverses across the domain gap. The ResNet backbone is more accurate on synthetic imagery (0.271 m vs the Transformer's 0.366 m). However, ResNet performs significantly worse on real imagery. CNNs degradation factor across domain is massive, 14.5 and 15.4; while the transformer is much more robust, degrading by factors of only 5.7 and 4.4. The mechanism behind this failure is visible in the predicted range distributions: on both hardware-in-the-loop domains, the ResNet-18 collapses onto a narrow peak near 2.2 m and returns nearly the same answer regardless of image content, while the transformer successfully retains a realistic spread.

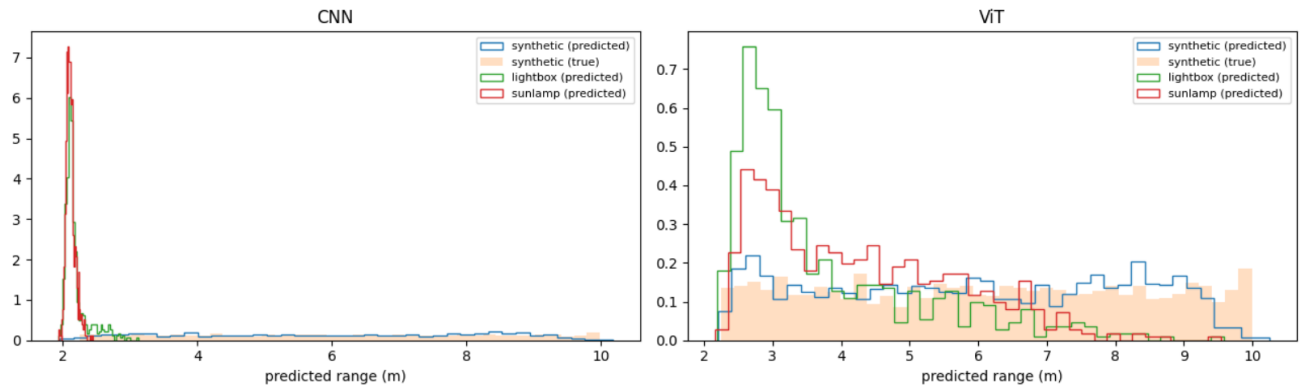


FIGURE 2: the collapse of the ResNet-18 to a near-constant prediction on real imagery, against the retained spread of the transformer DINOv2

Error is strongly anisotropic. Both models struggle significantly more with depth than with lateral positioning. For the ResNet-18, the median error along the camera's boresight (depth) is 0.258 m compared to just 0.061 m across it (lateral), a factor of 4.2. The transformer shows a similar but less extreme trend (0.297 m vs. 0.138 m, a factor of 2.2). This discrepancy occurs because a single camera easily determines lateral position based on where the target appears in the frame, but must infer range purely from how large it appears. This observability problem and argument is the exact reason why the cooperative architecture in [1] relies on a laser rangefinder to measure range.

The necessity of domain adaptation. Neither model is usable on real, untreated imagery. Both exhibit severe degradation, with position errors reaching 26% to 67% of the total range and rotational errors exceeding 100°. This aligns with the SPEC2021 analysis [5], which concluded that competitive solutions heavily depended on domain adaptation using target-domain imagery.

Training cost vs. trainable parameters. having fewer *trainable* parameters did not equate to faster training. Even though the transformer's backbone was frozen, it took 42 seconds per epoch compared to the fully fine-tuned ResNet's 24 seconds. Freezing weights removes the need for a backward pass through the backbone, but the computationally expensive forward pass remains unchanged.

5. Experiment 5: aleatoric and epistemic uncertainty

Idea. The confidence signal in Experiment 3 is generated by a solver and has no units. But, a navigation filter requires a calibrated covariance matrix. This experiment trains a network that predicts its own error and separates it into two components: aleatoric uncertainty, which is noise in the observation and is predicted by a variance output trained with a Gaussian negative log-likelihood [10]; and epistemic uncertainty, the model's own ignorance, estimated by Monte Carlo dropout [11]. Epistemic uncertainty requires no labels, so it is a candidate mechanism for detecting a domain shift in flight.

Result on synthetic domain. Predicted aleatoric standard deviation is [0.097, 0.121, 0.906] m per axis and the epistemic uncertainty is [0.048, 0.069, 0.670] m. The boresight (depth) component is seven to nine times larger than the lateral components. The model learned this physical reality without direct supervision on that split, and it is consistent with the 4.2x error anisotropy measured directly in Experiment 4. The model's calibration is conservative: the median ratio of actual error to predicted standard deviation is 0.38, meaning the predicted error bars are wider than the observed errors. The correlation between actual error and predicted standard deviation is +0.304.

Domain	Aleatoric (m)	Epistemic (m)	Epistemic ratio	ROC area
synthetic	0.375	0.262	1.00	—
lightbox	0.135	0.170	0.65	0.251
sunlamp	0.140	0.171	0.65	0.256

Table 3. Predicted uncertainty by domain, and ROC area for separating synthetic from hardware-in-the-loop images using the epistemic term alone.

Result across the Domain gap: the core hypothesis is falsified. Epistemic uncertainty does not rise on out-of-distribution domains. Instead, it falls to roughly 65% of its synthetic baseline. When scored as an out-of-distribution detector, the areas under the ROC curve are 0.251 and 0.256. Because these values fall well below 0.5 (the baseline of an uninformative, random signal), it indicates the relationship is systematically inverted. The model is measurably more confident on the two domains where Experiment 4 measured its actual error to be 14.5 and 15.4 times larger.

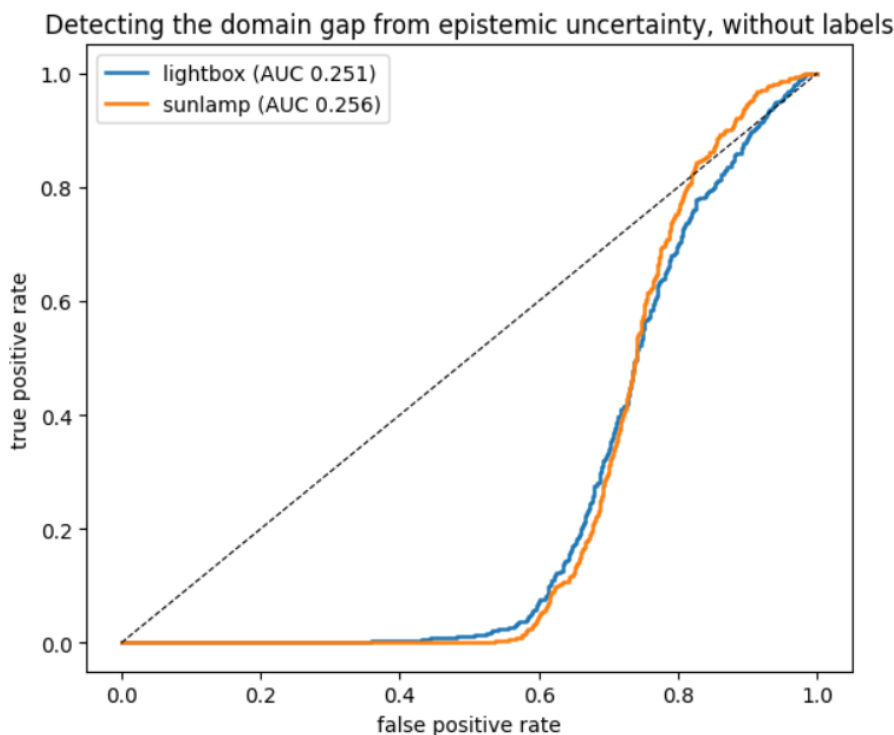


FIGURE 3: ROC curves falling below the diagonal, an area of approximately 0.25 rather than the 0.5 of an uninformative signal

Mechanism. Monte Carlo dropout estimates uncertainty by measuring the disagreement between multiple stochastic forward passes. Experiment 4 demonstrated that on real imagery, the network's predictions collapse to a near-constant output. Because the network outputs the same collapsed prediction every time, the stochastic passes perfectly agree with one another, resulting in an artificially low epistemic estimate. **Prediction collapse presents as confidence.** While an ROC area of 0.25 means the signal becomes mathematically informative if inverted, an inverted detector cannot be trusted on a domain against which it has not been calibrated, which is the only scenario where it would be useful.

6. Experiment 6: does a confidence signal help the navigation filter?

Idea. Experiments 1 to 5 evaluate performance on single, static images. However a navigation system runs a filter over a sequential trajectory. What a navigation filter requires from perception system is, a measurement together with a measurement covariance R , which determines how aggressively the state estimate moves towards each new measurement. A confidence signal is only useful if it can dynamically and accurately set R . To test this, the dynamics, filter structure, campaign style, and reporting format follow the cooperative rendezvous architecture detailed in [1].

Method. The simulation models a V-bar approach from 20 m using Hill-Clohessy-Wiltshire (HCW) relative dynamics; Measurements are produced by running the Experiment 1 pose chain on rendered frames at 1 Hz. These feed an Extended Kalman Filter tracking relative position and velocity; The experiment consists of 60 Monte Carlo runs of 100 steps with randomised degradation profiles. Filter consistency is scored using the **Normalized Estimation Error Squared (NEES)**, which for a three-axis position state should sit near 3.0.

The symmetry problem (a geometric failure, not a tuning artefact). The target spacecraft is exactly symmetric under a 180° rotation about its z axis. Mathematically, rotating the spacecraft 180 degrees maps the keypoints perfectly back onto themselves with zero residual error. No appearance-based method can resolve this visual ambiguity from a single image. More importantly, this failure was generated by the actual pose solver rather than an artificial error model, proving that the algorithm genuinely reports high confidence even when the orientation is completely flipped.

	Correct association	Flipped association	Difference
Median position error	0.069 m	0.069 m	none
Median attitude error	0.76°	179.9°	179°
Mean inlier ratio	0.908	0.906	0.002

Table 4. The symmetry failure corrupts attitude while leaving position and confidence unchanged. 90% of flipped frames still report confidence above 0.8.

Configuration	RMSE (m)	Final error (m)	Median NEES	Used
Fixed R	0.434	0.039	8.88	79%
Adaptive R	0.432	0.035	9.17	79%
Gated (floor 0.35)	0.432	0.035	9.17	79%

Table 5. Monte Carlo campaign, 60 runs. Expected NEES for a consistent filter is 3.

Result. The confidence signal fails to distinguish a 180° attitude error from a correct solution (an inlier ratio of 0.906 vs. 0.908). This exposes a fundamental limitation: a geometric confidence signal reports how well a hypothesis was fitted, not whether the correct hypothesis was fitted.

Because of this, setting R per frame from that signal gives no improvement over using a single fixed value. The gated configuration is identical to the adaptive one because the gate never fired, no measurement fell below the 0.35 confidence floor. Thus, effectively only two configurations were tested, not three.

Finally, the filter is overconfident in every configuration, yielding a NEES near 9.0 rather than the expected 3.0. This indicates that the process noise and the measurement model itself require correction before tweaking the R schedule, That will yield benefits. Corroborating the findings of Experiments 4 and 5, the measurement covariance once again shows that the range axis is roughly eight times worse than the lateral axes.

7. Experiment 7: deployment cost

Method. Accuracy alone does not decide model deployment. Rendezvous navigation runs entirely onboard the spacecraft, since round-trip communication to ground control takes seconds at best, and is often unavailable during the critical minutes it takes to traverse the docking corridor. Furthermore, representative space-flight processors are radiation-hardened and have no GPU. A modern CPU is two to three orders of magnitude faster than a GR740, so the single-thread CPU figure below should be viewed as an optimistic proxy. Thus, achieving flight-rate inference usually requires dedicated hardware acceleration [12][13] rather than simply changing the software architecture.

Model	Params	Size	GMACs	GPU	CPU 1-thread
ResNet-18	11.57 M	46.3 MB	1.74	3.6 ms	48.9 ms
DINOv2 ViT-S/14	22.58 M	90.4 MB	6.12	7.4 ms	175.9 ms

Table 6. Per-frame cost, batch size 1, warm-up discarded and CUDA synchronised.

Result. For a single forward pass, both models fit comfortably within a standard time budget: Assuming a 1 Hz camera framerate where half the time (500 ms) is allocated to perception, the convolutional model uses only 10% of the budget (48.9 ms), and the transformer uses 35% (175.9 ms).

However, the uncertainty estimate is what breaks the system. Running the Twenty-five stochastic Monte Carlo passes(required for Epistemic uncertainty) take 1223 ms for the ResNet and 4397 ms for the transformer. This exceeds the 500 ms perception budget by factors of 2.45 and 8.79, respectively. This hidden computational penalty is a major operational cost not reflected in the accuracy tables of Experiments 4 and 5.

Quantisation. Dynamic INT8 quantisation reduces the transformer’s memory footprint by a factor of 3.6 (from 90.4 MB down to 25.2 MB). But this compression introduces severe instability, shifting the predicted translation by up to 1.62 m, an error larger than the model's own baseline median error on synthetic data.

Conversely, the convolutional model quantises safely (predictions shift by a negligible 0.032 m) but barely shrinks, reducing by a factor of just 1.03. This happens because dynamic quantisation targets linear layers, whereas the ResNet’s arithmetic is almost entirely convolutional. Neither outcome is usable as it stands: deploying the transformer would require quantisation-aware training, while the convolutional model would require static quantisation of its convolutions.

8. Synthesis and Conclusion

Four confidence mechanisms were evaluated across two targets and two datasets. All four fail in the same direction: **they report high confidence where the system is most wrong.**

Exp.	Mechanism	Observed failure
3	RANSAC inlier ratio	Learned detector worst calibrated of three, ECE 0.232 against 0.091
4	Predicted range distribution	Collapses to a near-constant value on real imagery; error rises 14.5x and 15.4x
5	Epistemic uncertainty	Falls to 0.65x where error rises 15x; ROC area 0.251 and 0.256, inverted
6	Inlier ratio inside an EKF	Identical on frames wrong by 180° in attitude; filter NEES 8.9-9.2 against 3.0

Table 7. Observed failures of each method

None of these four failures are mere tuning artefacts. As demonstrated in Experiment 6, the EKF failure is driven by a mathematically proven geometric symmetry. In Experiment 5, the epistemic failure is a direct consequence of prediction collapse, which Experiment 4 measured independently on real imagery.

The Observability Problem:

Beyond confidence estimation, a consistent secondary finding emerges regarding depth observability.

The along-boresight(range) axis is identified as the system's weakest link three separate times:

- (A) It exhibited a 4.2x worse error than the lateral axes on the SPEED+ dataset (Experiment 4).
- (B) It was predicted to be seven to nine times larger in the model's own uncertainty estimates (Experiment 5).
- (C) It was roughly eight times larger in the navigation filter's measurement covariance (Experiment 6).

Because two of these three confirmations were measured on real, hardware-in-the-loop imagery, this is no longer just a theoretical limitation. This data empirically validates the well-known observability argument: to safely determine range, cooperative rendezvous architectures must rely on a laser rangefinder rather than a monocular camera.

9. Limitations of this study:

- **Training Budget:** Training used 8,000 of 47,966 available SPEED+ synthetic images for 12 to 15 epochs. Absolute errors are therefore quite high, and well above published SPEC2021 competition leaderboards; The objective here was to compare two backbones under an identical, constrained computational budget.
- **Direct Regression:** Attitude is regressed directly as a quaternion, with no keypoint or geometric stage. This is the weaker of the two standard approaches and shows in the rotation errors.
- **No domain adaptation:** Domain adaptation was deliberately omitted to ensure the untreated domain gap was what was actually being measured and reported.
- **Asymmetric Backbone Training:** The transformer backbone is frozen, so the comparison is a fine-tuned convolutional network against a frozen self-supervised representation with a trained head.
- **Uncertainty Estimation Limits:** Monte Carlo dropout is a cheap approximation to a posterior, and dropout is present only in the network head. A deep ensemble is the stronger estimator, so the negative results in Experiment 5 do not answer the question for epistemic uncertainty in general.
- **Simulated Trajectories:** Experiment 6 uses a simulated trajectory and rendered imagery, since no public dataset provides labelled rendezvous sequences at the scale a Monte Carlo campaign requires.
- **Hardware Proxies:** Deployment figures come from general-purpose hardware. Power and radiation tolerance are not measured.

10. Proposed direction

These experiments explore possible solutions to develop a failure detection mechanism for learned pose estimation that remains trustworthy. The following are the possible directions that can be explored to solve this problem.

- **Apply domain adaptation** and establish whether uncertainty behaviour improves once the gap is treated rather than merely observed, which is the condition under which a learned front end could be considered for a filter.
- **Replace Monte Carlo dropout with a deep ensemble** and test whether the inversion observed in Experiment 5 persists, or whether it is specific to dropout applied to the head alone.
- **Adopt keypoint regression followed by PnP** instead of direct pose regression, so that geometric consistency provides a check independent of the network's own confidence. This is the architecture that performed good in SPEC2019 [3] competition, and the one closest to the classical chain already in operational use [1].

References

- [1] A. Martinez, J. Ramirez, F. Cacciatore, P. Gonzalez, "Guidance, Navigation and Control for the autonomous rendezvous and docking of cooperative targets", ESA GNC-ICATT, 2023.
- [2] S. Sharma, C. Beierle, S. D'Amico, "Pose estimation for non-cooperative spacecraft rendezvous using convolutional neural networks", IEEE Aerospace Conference, 2018.
- [3] M. Kisantal, S. Sharma, T. H. Park, D. Izzo, M. Märtens, S. D'Amico, "Satellite pose estimation challenge: dataset, competition design, and results", IEEE Trans. Aerospace and Electronic Systems, vol. 56, 2020.
- [4] T. H. Park, M. Märtens, G. Lecuyer, D. Izzo, S. D'Amico, "SPEED+: next-generation dataset for spacecraft pose estimation across domain gap", IEEE Aerospace Conference, 2022.
- [5] T. H. Park et al., "Satellite pose estimation competition 2021: results and analyses", Acta Astronautica, vol. 204, 2023.
- [6] L. Pauly, W. Rharbaoui, C. Shneider, A. Rathinam, V. Gaudillière, D. Aouada, "A survey on deep learning-based monocular spacecraft pose estimation", Acta Astronautica, vol. 212, 2023.
- [7] M. Tyszkiewicz, P. Trulls, V. Lepetit, P. Fua, "DISK: learning local features with policy gradient", NeurIPS, 2020.
- [8] C. Guo, G. Pleiss, Y. Sun, K. Q. Weinberger, "On calibration of modern neural networks", ICML, 2017.
- [9] M. Oquab et al., "DINOv2: learning robust visual features without supervision", TMLR, 2024.
- [10] A. Kendall, Y. Gal, "What uncertainties do we need in Bayesian deep learning for computer vision?", NeurIPS, 2017.
- [11] Y. Gal, Z. Ghahramani, "Dropout as a Bayesian approximation", ICML, 2016.
- [12] K. Cosmas, K. Asami, "Utilization of FPGA for onboard inference of landmark localization in CNN-based spacecraft pose estimation", Aerospace, vol. 7, 2020.
- [13] A. Posso, G. Bois, Y. Savaria, "Mixed-precision quantization for spacecraft pose estimation on reconfigurable MPSoC", IEEE NEWCAS, 2024.